No constant pole parameter is complete

Abstract

We consider the two-equation pole certificate used in the rational-source construction of the integers. No fixed rational value of its parameter e is complete on the prescribed set of prepared inputs. For either degenerate value $e = 1, -1/3$, the same prepared input $b = (204/55)^2$ gives a genus-one quartic with points over every completion of $\mathbb{Q}$ and no rational point. Exact unconditional two-descent and Cassels-pairing computations certify the two elliptic rank bounds used in the proof. An elementary squareclass argument excludes every other constant value of e. The obstruction includes zero values of the second square root, so removing a nonzero guard does not repair it.

1 Prepared inputs and the pole equations

Definition 1.1. A rational number b is *prepared* if its reduced denominator is a square and

$$0 < b < 17, \qquad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2},$$
$$v_2(b) \geq 3, \qquad v_3(b) \geq 1, \qquad v_{17}(b) \geq 2.$$

For such b, put $B = 17b$ and $V = B(B - b^2)$. The pole equations are

$$p^2 = B(1 - e)(1 + 3e), \qquad Bq^2 = (s^2 - eV)(s^2 + 3eV). \tag{1}$$

All four witnesses are rational; zero values are allowed.

These are exactly the preparation hypotheses and the zero-preparation equations stated in [1, Section 3]. Its realization theorem permits e to vary. Our question is whether a fixed value can replace that choice.

Theorem 1.2. *For every $e_0 \in \mathbb{Q}$, there is a prepared b for which (1) has no rational solution with $e = e_0$. For $e_0 = 1$ and $e_0 = -1/3$, one can take*

$$b_0 = \frac{41616}{3025} = \left(\frac{204}{55}\right)^2.$$

For either of these two values, the second equation has no rational solution even though its smooth projective quartic has points over every completion of $\mathbb{Q}$.

The rank bounds in the proof are exact computations. They use the unconditional algebraic rank bounds of PARI/GP, not an analytic-rank calculation or the Birch–Swinnerton-Dyer conjecture. The complete verifier and its exact intermediate data accompany this note.

2 Two elliptic rank certificates

For $n \in \mathbb{Q}^\times$, write

$$E_n : \quad y^2 = x(x - n)(x + 3n).$$

These curves have the three rational nonidentity two-torsion points $(0, 0), (n, 0), (-3n, 0)$.

Proposition 2.1 (Exact computation)**.** *The groups $E_{-29427}(\mathbb{Q})$ and $E_{9809}(\mathbb{Q})$ have ranks zero and one, respectively. Both have torsion subgroup $(\mathbb{Z}/2\mathbb{Z})^2$. The point*

$$P_* = (-20449, 2356926) \in E_{9809}(\mathbb{Q})$$

has infinite order.

Proof. The verifier initializes the two curves with coefficient vectors $[0, 2n, 0, -3n^2, 0]$. PARI/GP version 2.17.2 returns

n	`ellrank` output
-29427	$[0, 0, 2, []]$
9809	$[1, 1, 2, [[-20449, 2356926]]]$.

Here the first two entries are unconditional lower and upper bounds for the rational rank. The algorithm computes the two-Selmer rank, the rational two-torsion rank, and the rank of the relevant Cassels pairing; the upper bound is the first rank minus the other two [2]. The saved exact diagnostics give two-Selmer ranks $4, 5$ and pairing matrices

$$C_- = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \qquad C_+ = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Independent row reduction over $\mathbb{F}_2$ verifies that each has rank two. Since the rational two-torsion has rank two, the certified upper bounds are $4 - 2 - 2 = 0$ and $5 - 2 - 2 = 1$.

For an independent torsion check, direct finite-field counting gives

n	$\#E_n(\mathbb{F}_5)$	$\#E_n(\mathbb{F}_7)$	$\#E_n(\mathbb{F}_{11})$	$\#E_n(\mathbb{F}_{13})$
-29427	4	8	8	12
9809	8	8	16	12.

All these primes are good. The injection on prime-to-p torsion under good reduction, using the different primes in the table, bounds the rational torsion order by four in each case. The known two-torsion attains this bound. Direct substitution verifies P_*, and its nonzero ordinate shows that it is not torsion. This also independently supplies the lower rank bound one for E_{9809}. □

Lemma 2.2. *For a nonsingular curve $y^2 = x(x^2 + ax + c)$ with $c \neq 0$, the map*

$$\alpha : E(\mathbb{Q}) \longrightarrow \mathbb{Q}^\times/\mathbb{Q}^{\times 2}, \qquad \alpha(O) = 1, \quad \alpha((0,0)) = c, \quad \alpha((x,y)) = x \quad (x \neq 0)$$

is a group homomorphism. It kills $2E(\mathbb{Q})$. In particular, if E has rank r and full rational two-torsion, its image has size at most 2^{r+2}.

Proof. Intersect a nonvertical line $y = ux + v$ with the curve. The product of its three intersection abscissas, counted with multiplicity, is v^2. Since reflection in the x-axis does not change an abscissa, this proves the homomorphism identity when $v \neq 0$. If $v = 0$, the two nonzero intersection abscissas have product c, which gives the identity when the third point is $(0, 0)$. Addition by $(0, 0)$ sends a nonzero abscissa x to c/x; vertical lines and two-torsion doubling give the remaining cases. The target has exponent two, so $2E(\mathbb{Q})$ is killed. Finally the Mordell–Weil decomposition gives $|E(\mathbb{Q})/2E(\mathbb{Q})| = 2^r|E(\mathbb{Q})[2]| = 2^{r+2}$. □

Corollary 2.3. *Neither $E_{-29427}(\mathbb{Q})$ nor $E_{9809}(\mathbb{Q})$ has a nonzero abscissa in the rational squareclass of* 17.

Proof. The first curve has only its four torsion points. Its nonzero abscissas are -29427 and $88281 = 9 \cdot 17 \cdot 577$; neither has squareclass 17.

On E_{9809}, the two-torsion points give squareclasses -3 and 9809, while P_* gives -1, because $20449 = 143^2$. These three classes are independent: $9809 = 17 \cdot 577$ and the prime 577 distinguishes the third generator from the first two. They therefore supply eight classes. By Lemma 2.2 and rank one, these exhaust the image:

$$\alpha(E_{9809}(\mathbb{Q})) = \{\pm 1,\ \pm 3,\ \pm 9809,\ \pm 29427\}.$$

The class 17 is absent. □

3 The two degenerate specializations

Put $M = 55$, $t = 204/M$, and $d = M^2 - 2448 = 577$. Then

$$b_0 = t^2, \qquad B = 17t^2, \qquad 17 - t^2 = \frac{17d}{M^2}, \qquad V = \frac{289t^4 d}{M^2}.$$

For $e = 1$, the invertible rational substitutions

$$s = \frac{17t^2}{M}u, \qquad q = \frac{17t^3}{M^2}v$$

turn the second pole equation into

$$C_+ : \quad v^2 = 17(u^2 - d)(u^2 + 3d). \tag{2}$$

For $e = -1/3$, the substitutions

$$s = \frac{17t^2}{3M}u, \qquad q = \frac{17t^3}{9M^2}v$$

give

$$C_- : \quad v^2 = 17(u^2 + 3d)(u^2 - 9d). \tag{3}$$

The verifier checks these identities coefficient by coefficient.

In either equation $u = 0$ makes the right side negative. Every rational affine point would therefore have $u \neq 0$. The map

$$(x, y) = (17u^2, 17uv)$$

sends C_+ to $E_{17d} = E_{9809}$ and C_- to $E_{-51d} = E_{-29427}$. Its nonzero abscissa has squareclass 17, contradicting Corollary 2.3. This proves that both quartics have no rational affine point. Their points at infinity would require 17 to be a rational square, so their smooth projective completions have no rational point either.

Proposition 3.1. *Both C_+ and C_- have points over every completion of $\mathbb{Q}$.*

Proof. Let $g_\pm(u)$ denote the corresponding right side. The following table gives rational values of u for which $g_\pm(u)$ is a nonzero square in the indicated local field. Each entry records u, the valuation, and the residue of the unit after removing that valuation. The unit is taken modulo 8 at 2 and modulo p at an odd prime.

p	$C_+ : (u, v_p(g_+), \text{unit})$	$C_- : (u, v_p(g_-), \text{unit})$
2	$(23, 6, 1)$	$(21, 6, 1)$
3	$(1, 2, 1)$	$(12, 4, 1)$
17	$(4, 2, 13)$	$(22, 2, 9)$
577	$(1, 0, 17)$	$(1, 0, 17)$.

The units $13, 9$ are squares modulo 17, and 17 is a square modulo 577. The standard local square criterion proves these four local assertions. At the real place, $u = 100$ makes both right sides positive.

The quartic discriminants are

$$\operatorname{disc}(g_+) = -2^{12}3\,17^6 d^6, \qquad \operatorname{disc}(g_-) = -2^{12}3^7\,17^6 d^6.$$

Thus the smooth projective genus-one curves have good reduction outside $2, 3, 17, 577$. At any other prime, the Hasse bound gives a point on the reduction, and smooth lifting gives a local point. This proves the proposition. □

4 Exclusion of all other constant values

The number $b_0 = (204/55)^2$ is prepared: its reduced denominator is square, it lies in $(0, 17)$, and its valuations at $2, 3, 17$ are $4, 2, 2$. Being a rational square, it satisfies all three local square conditions.

A second prepared number is

$$b_1 = 409\left(\frac{204}{1001}\right)^2 = \frac{17020944}{1002001}.$$

It is less than 17 because $17020944 < 17034017$. Its denominator is square and its valuations at $2, 3, 17$ are again $4, 2, 2$. The number 409 is 1 modulo $8, 3, 17$, so it is a square in all three local fields.

Suppose $e_0 \neq 1, -1/3$ made the first pole equation soluble for both inputs. Its factor $(1 - e_0)(1 + 3e_0)$ is nonzero; division of the two equations would make b_1/b_0 a rational square. But

$$\frac{b_1}{b_0} = 409\left(\frac{55}{1001}\right)^2$$

is not a square. Thus at least one of the two prepared inputs already fails the first equation. The two excluded degenerate values fail at b_0 by Section 3. This completes the proof of Theorem 1.2.

The theorem concerns a constant e. It does not exclude a nonconstant function of b, a different relation between the pole parameters, or a different pole predicate. Those proposals require their own completeness arguments.

Reproducible verification

Run `check_fixed_e_obstruction.py` to reproduce `fixed-e-obstruction-certificate.json` and the descent transcript `fixed-e-pari-descent.txt`. The script independently checks preparation, substitutions, torsion, squareclasses, local witnesses, and pairing-matrix ranks using exact integer and rational arithmetic. The two algebraic rank calculations use PARI; no bounded search is used as a nonexistence certificate.

References

[1] *Arithmetic dependencies of the construction with 29 variables*, project paper, version 8, 8 November 2026.

[2] The PARI Group, *PARI/GP: Elliptic curves*, documentation of `ellrank`, including its unconditional two-descent and Cassels-pairing bounds, `https://pari.math.u-bordeaux.fr/dochtml/html/Elliptic_curves.html#ellrank`.